

Introduction to Computer Programming I CSCI 1010-03 – Fall 2025

Course Information

Course Description

A first course in computer programming. Students will use a high-level object-oriented programming language (Java) to design, implement, debug, and test interactive software. In addition to programming fundamentals, students will gain early exposure to concepts of computational efficiency, scalability, and high-performance computing (HPC). Through hands-on labs and enrichment activities, students will explore how programs behave as input size increases, how parallelism can improve performance, and how supercomputing resources are applied to solve real-world problems.

- Prerequisites: none
- Corequisites: CSCI 1011
- Credit Hours: 3
- Required Textbook: Java: An Introduction to Problem Solving and Programming, 8th edition by Walter Savitch (ISBN: 978-0-13-446203-5)
- Time and Place: MWF 9:05 am – 10:00 am, Maynard 129

Instructor Information

- Instructor Name: Dr. Sabrina Perry
- Office: Maynard 202
- Office Hours:
In-person
Maynard 202 Mon 8:00 am – 9:00 am Tue 9:00 am – 11:00 am; Wed 3:00 pm – 5:00 pm
or by appt.
Office Hours Format: Microsoft Teams. New Chat “perrys1@apsu.edu”, followed by video call if needed.
- Phone: 931-221-7416
- Email: perrys1@apsu.edu

Programming Tutoring

- Check out programming peer tutoring service/timing/venue at **this link**, or use the following URL:
- **https://www.apsu.edu/csci/opportunities_resources/computer-science-tutoring-services.php**

Course Description

A first course in computer programming. Students will use a high-level object-oriented programming language to design, implement, debug, and test interactive software.

Course Objectives

At the completion of the course, the student will be able to:

- Use a programming language to implement, test, and debug algorithms for solving simple problems.
- Design, implement, test, and debug programs that use each of the following fundamental programming constructs: basic computation, simple I/O, standard conditional and iterative structures, the definition of functions, and parameter passing.
- Trace the execution of a variety of code segments and write summaries of their computations.
- Explain why the creation of correct program components is important in the production of high-quality software.
- Conduct a personal code review (focused on common coding errors) on a program component using a provided checklist.
- Construct and debug programs using the standard libraries available with a chosen programming language.
- Apply consistent documentation and program style standards that contribute to the readability and maintainability of software.
- Recognize how computational problems scale with input size and system resources.
- Demonstrate introductory knowledge of HPC concepts, including parallel execution and supercomputing applications.

Student Learning Outcomes

- Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions. (SLO1)
- Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline. (SLO2)
- Support the delivery, use, and management of information systems within an information systems environment. (SLO6-CIS)
- Use systemic approaches to select, develop, apply, integrate, and administer secure computing technologies to accomplish user goals. (SLO6-CIT)
- Explain the difference between sequential and parallel execution in a computing context.
- Identify real-world applications of supercomputing and how programming fundamentals contribute to solving large-scale problems.

Time and Place of Class:

- CON – Face to Face Instructions
MWF 9:05 am – 10:00 am, Maynard 129

Learning Resources

This course uses D2L. The syllabus, lecture notes and assignments will be found via the [D2L login page](#). The student should visit this site at least once daily. All communication from the instructor to the students in the course can be found in the **Announcements** section in D2L. **Do not use D2L's email system for this course.** Send all email to perrys1@apsu.edu from your APSU email account. **Do not email the instructor using an external account.**

Minimum Technology Requirements

- Consult the [CSIT Department computer and software recommendations for coursework](#) for the minimum technology requirements for all courses. The following software will be used in this course:

Java: Download from <https://adoptium.net/> (JDK 24)

Java IDE: Apache NetBeans download from
<https://netbeans.apache.org/download/index.html>

- Laptop Computer with webcam, and mouse.
- Everyday Internet access capable of supporting Zoom video conferencing.
- **Zoom Client**
 - <https://www.apsu.edu/online/technology/zoom/index.php>
 - Login to Zoom
 - Install Client Software
 - **sign in with SSO**
 - **apsu.zoom.us** (company domain)
 - use apsu email login, you should see “licensed” next to your name
- **Microsoft Teams Client**
 - MS Teams is part of MS Office 365.
 - APSU students can install Office 365 Apps on up to 5 personal devices, FREE.
 - To install Office 365 Client
 - <https://govstech.apsu.edu/TDClient/2071/Portal/Requests/ServiceDesk?ID=14541>
 - Install Office 365
 - Login in using apsu login
 - After login to outlook, upper left corner (Icon) > Office > Install Office

- Install MS Teams Client, login in with apsu login

Course Policies

Attendance

Students are expected to attend all classes, arrive on time and participate in classroom discussions. Students are expected to remain in class until the class is finished unless extenuating circumstances such as illness require the student to leave. If a student leaves class early, the instructor must be notified as to the reason that the student left.

A grade of FA (to include the last date of attendance) will be reported within 14 days of the last day of attendance. A grade of FN and an entry of the first date of class in the last date of attendance column will be reported within the first 14 days of class. The FN grade is considered to be an official withdrawal from the class and may impact the enrollment status of a student (full-time to part-time).

Assignments

Assignments (projects) will be both written and programming. The instructor will not make any adjustments to a student's code when grading, so if any submitted program does not run the student will get a zero on the "correctness" portion of the grade (or **50 points off** the overall grade), with no exceptions. Program source code will be turned in electronically.

In addition to programming projects, some assignments may involve guided use of generative AI tools. Students will be required to reflect on the output, compare it to their own work, and demonstrate understanding of the underlying programming concepts.

Late Policy for Assignments

Assignments (projects) are due on the stipulated due dates, and may be turned in **up to 7** calendar days late with a **25% late penalty**. No assignment will be accepted more than 7 calendar days after the original due date! Students with planned absences, whether for university events, religious observance, or other reason, are expected to make arrangements with the instructor to turn in assignments or take exams before the scheduled date of the assignment or test.

Days Late	Penalty
0 to 7	-25% of assignment score
> 7	-100% (0) OR discuss with instructor

Exams

Exam/Quizzes dates are on the schedule on the following page — if there are any changes, they will be announced at least one week in advance if possible. A missed exam may be made up only if it was missed due to an extreme emergency and arrangements are made before the exam date. Exams may not be taken early or late due to personal travel plans.

Grades

The following weights will be used in assigning a grade for this course:

Activity	Percentage
Quizzes	20%
Assignments	40%
Midterm Exam	20%
Final Exam	20%
AI/Enrichment Activities	Up to 5% bonus credit

Midterm and final grades will be assigned as follows:

Percentage	Grade
90-100%	A
80-89%	B
70-79%	C
60-69%	D
0-59%	F

Midterm Grades

A midterm grade will be awarded for all students in this course. The grade awarded may not necessarily be based on 50% of the course requirements and may or may not differ from the final grade. Your midterm grade will be posted online. Any grade discrepancy must be submitted in writing, along with original document, within one week of distribution. Notice that the grade breakdowns are given in percentages, not points.

General Policies

Policy on Minors

Minors accompanying staff, faculty, students, or visitors on campus are not permitted in classrooms. However, affiliated minors may utilize classrooms designated for use by a program approved by the university in which they are a participant.

Service Animals in the Classroom

Consult [Policy 3:007 Animals on Campus](#) for appropriate situations allowing service animals in the classroom.

Students with Disabilities

Any student who has a disability that may affect his/her academic performance is encouraged to make an appointment with me to discuss this matter, or you may contact Disability Services; telephone 221-6230; tty 221-6278; fax 221-7102.

Academic and Classroom Misconduct

Students are expected to conduct themselves appropriately at all times. Academic and classroom misconduct will not be tolerated. Students must read the “Code of Student Conduct” in the new Student Handbook for an understanding of what will be expected of them within the academic setting. [Policy 3:005 Student Academic and Classroom Misconduct](#) will be followed in reporting any suspected cases of academic misconduct.

Students are required to turn in their own work and not the work of others. Collaboration on homework, assignments, quizzes, and exams is prohibited, unless otherwise specified by the instructor. Likewise, plagiarism of other’s work or web-related sources constitutes a serious infraction. This includes submitting work very similar to another student’s project, copy and paste from Internet searches, using code found on another computer, and using Internet sites for hiring coders or tutors to complete a project. “Penalties for academic misconduct will vary with the seriousness of the offense and may include, but not limited to, a grade of “F” on the work in question, a grade of “F” in the course, reprimand, probation, suspension and expulsion.” (Quoted from APSU Academic and Classroom Misconduct). Protect your own work. Do not leave your assignments on the hard drives of the computers or printers in the lab for others to see. If you are ever worried that anything you do may be flagged for academic dishonesty, please contact me for help and guidance.

Laptops and Electronic Devices in the Classroom

Cell phones must be turned off or on vibrate during class. NO cell phone can be answered during class unless there is an emergency situation and you have discussed the emergency with the instructor prior to class. No text messaging is permitted during class.

Laptops/tablets are permitted whenever the student considers the laptop/tablet as an enhancement of his/her learning experience. However, the student **MUST** be using the laptop/tablet in a manner that directly relates to the content of this class such as viewing the slides from the lecture or taking notes. The laptop/tablet must not be a distraction to others in the class. No other electronic devices will be allowed in class without prior consent of the instructor.

Tentative Class Schedule

Week / Date	Topic	Reading	Remarks / Due / Exam
1 8/25, Mon	Introduction to Computers and Java	Book: 1.1-1.3, 2.1 Slides - Savitch 1.1-1.3	8/25: APSU Day of Service: No classes
2 9/1, Mon	Basic Computation: Variables & Expressions	Book: 2.1 Slides - Savitch 2.1 <i>*AI Reflection Prompt: "Ask an AI tool to explain what a Java variable is. Rewrite the explanation in your own words."</i>	9/1: Labor Day: No Classes 9/7: Last day to drop class w/out record Quiz 1 – 9/3, Wed
3 9/8	Basic Computation: Strings, I/O, & Programming Style	Book: 2.2-2.4 Slides - Savitch 2.2-2.4	Quiz 2 – 9/10, Wed
4 9/15	Flow of Control: Branching	Slides - Savitch 3.1 <i>*Introduce runtime measurement – students run the same loops with small vs. large input to see performance differences.</i>	Assignment 1 Due – 9/15 Mon Quiz 3 – 9/19, Fri
5 9/22	More Branching and Loops	Book: 3.2-3.3, 4.1 Slides - Savitch 3.2-3.3 Slides - Savitch 4.1	Quiz 4 – 9/26, Fri 9/22: Priority Advising and Registration begins for the Spring and Summer Semester 2026 9/25: Last day to drop with an automatic grade of 'W'
6 9/29	More Loops	Book: 4.2 Slides - Savitch 4.2 <i>*AI Debugging Activity: Students write a buggy loop, then compare their fix to AI's suggestion.</i>	Assignment 2 Due – 9/29, Mon Quiz 5 – 10/3, Fri
7 10/6	Mid-Term Review	Mid-Term Review <i>*Mini-activity: Cornhole with one hand vs. two hands" - discussion of sequential vs. parallel work.</i>	10/6-10: Registration begins for the Spring and Summer Semester 2026 Quiz 6 – 10/10, Fri Mid-Term Exam – 10/10 Fri

8 10/13			10/13 - 14: Fall Break, No classes
9 10/20	Defining Classes and Methods	Book: 5.1-5.2 Slides - Savitch 5.1 Slides - Savitch 5.2	Assignment 3 Due – 10/20 Mon

Week / Date	Topic	Reading	Remarks / Due / Exam
10 10/27	Defining Classes and Methods	Book: 5.3 Slides - Savitch 5.3	Quiz 7 – 10/31, Fri
11 11/3	More About Objects and Methods	Book: 6.1-6.2 Slides - Savitch 6.1-6.2	Quiz 8 – 11/7, Fri
12 11/10	More About Objects and Methods	Book: 6.3-6.7 Slides - Savitch 6.3-6.7	11/11: Veteran's Day: No Classes Assignment 4 Due – 11/10, Mon Quiz 9 – 11/14, Fri
13 11/17	Arrays	Book: 7.1-7.2 Slides – Savitch 7.1-7.2 <i>*Discuss scalability of arrays – “how do arrays behave when scaled to very large datasets?*</i>	Quiz 10 – 11/21, Fri
14 11/24	Exceptions, and Files	Book: 9.1, 10.1-10.2 Slides - Savitch 9.1 Slides - Savitch 10.1-10.2 <i>*Show how supercomputing systems manage large data files.*</i>	11/26 - 28: Thanksgiving Holiday: No Classes Assignment 5 Due – 11/24, Mon Quiz 11 – 11/24, Mon
15 12/1			Assignment 6 Due – 12/3, Wed Quiz 12 – 12/3, Wed Final Exam – 12/5 Fri for online classes (OR depending on class time for on-ground classes) 12/3, Wed: Last day of classes

			12/4, Thu: Study Day 5/5-11: Final Exam Period
16 12/8	Exam Revision		Final Exam - Depending on class time for on-ground classes

CAVEAT

The above schedule, policies, procedures, and assignments in this course are subject to change in the event of extenuating circumstances.

Syllabus Supplement

ZOOM Guidelines

Some or all of the class sessions may be audio/visually recorded. The video and/or audio recordings may be used for educational purposes and may be made available to all students currently enrolled in the course and to university officials with a legitimate educational interest in reviewing the recording. The recording is made for educational purposes to provide a resource to students during the remainder of the course.

Students should not make their own recording of the class session or distribute a copy of the instructor's recording of the class session unless appropriate approval has been received prior to making the recording. Distributed recordings are not a transfer of any copyrights in the recording. Public distribution of such materials by students may constitute copyright infringement in violation of federal or state law, or University policy. Further, the University prohibits the recording or transmission of classroom lectures and discussions by students unless written permission from the instructor has been obtained and all students in the class as well as guest speakers have been informed that audio/video recording may occur. Violators may be subject to disciplinary action pursuant to the Student Code of Conduct.

Statement from APSU Academic Affairs and Legal Affairs

APSU is committed to the free and full exchange of ideas and perspectives that is central to the educational enterprise. We are also committed to encouraging students—and all people—to be exposed to, and think critically about, sensitive topics and issues. This is an essential element of higher education and necessary to better prepare students for community participation and robust civic engagement. Curricular materials on concepts including but not limited to racism, sexism and classism may be presented and discussed in this class; while students are expected to master course content, it is not expected that students endorse or subscribe to any theory or viewpoint.

AI policy

AI-based assistance, such as ChatGPT and GitHub Copilot, will be treated the same way as collaboration with other people: you are welcome to talk about your ideas and work with others, both inside and outside the class, as well as with AI-based assistants. However, all work you submit must be your own. You should never include in your assignment anything that you yourself did not write. You should not copy/paste any code, text, image, etc., generated by an AI system and submit it as your own work. Anything found in your assignment that you did not personally write, whether from another student or an AI-based system, will be treated as an academic misconduct case.

In this course, you will also encounter structured AI activities designed to build your AI literacy and programming skills. For example, you may be asked to use an AI tool to

explain code, compare debugging approaches, or reflect on performance differences. These activities are learning opportunities, not shortcuts. Students will be evaluated on their ability to explain AI-generated content in their own words and to demonstrate understanding of the underlying programming concepts.

If AI use is suspected in your submitted work, you will not automatically be accused of academic misconduct. Instead, you may be asked to explain your code and reasoning. Students who demonstrate understanding and knowledge of the material may be considered not to have committed academic misconduct.

HPC and Emerging Technologies

This course introduces students to foundational ideas in high-performance computing (HPC), including parallelism, concurrency, and supercomputing. Students will explore how small Java programs scale with input size, gain exposure to classroom activities based on HPC Carpentry and Cornell Virtual Workshop modules, and learn how HPC powers real-world applications in science, engineering, and AI.